Cyber Power as a Structural Variable of Maritime Power

A conceptual update of its formulation — from a middle regional power's perspective

CF Diego Edison Cabuya Padilla, PhD · Naval Academy for Strategic Studies · Bogota, Colombia · 2026

THE PROBLEM

The **classical formula** of maritime power (Maritime Interests × Naval Power) does not capture the dynamics of **digitalized maritime environments**, where navigation, traffic, port facilities, and ocean surveillance depend on interconnected digital infrastructures vulnerable to cyberattacks.

FROM THE CLASSICAL TO THE EXTENDED FORMULA

$$MP = MI \times PN$$

CLASSICAL · 2 VARIABLES

→

$$MP_{norm} = [(MI \times NP) \times SW \times MS^{\gamma} \times CP^{\alpha}]^{1/n_{eff}}$$

EXTENDED · 5 VARIABLES

MI	NP	SW	MS	CP
Maritime Interests	Naval Power	Strategic Will	Maritime Security	Cyber Power

FIVE-LEVEL INTERPRETATION SCALE

Critical	Low	Moderate	Significant	High
0.00–0.20	0.21–0.40	0.41–0.60	0.61–0.80	0.81–1.00

THE EVIDENCE — COLOMBIA CASE STUDY

0.53	33/99	CP	3
Normalized maritime power (Moderate)	Global maritime power ranking	Most efficient investment vector	verification stages (properties·sensitivity·profiles)

FIVE ORIGINAL CONTRIBUTIONS

1. **Cyber power** formalized as a fifth structural variable of maritime power.
2. **Strategic will** repositioned as a multiplier of the complete system.
3. **Maritime security** disaggregated into three threat dimensions.
4. **Normalization** yielding an interpretable index in (0, 1].
5. **National Maritime Power Index (NMPI)** proposed as a comparative research agenda, with an open-access simulator.

Cyber Power as a Structural Variable of Maritime Power · Cabuya Padilla, D. E. (2026)
Naval Academy for Strategic Studies (ANEES) · Interactive simulator: github.com/diegocabuya/Cyber-Power-and-Maritime-Power